Project Title:

Parameterized Matrix Multiplier Accelerator with UVM Verification

Group No.- **08**

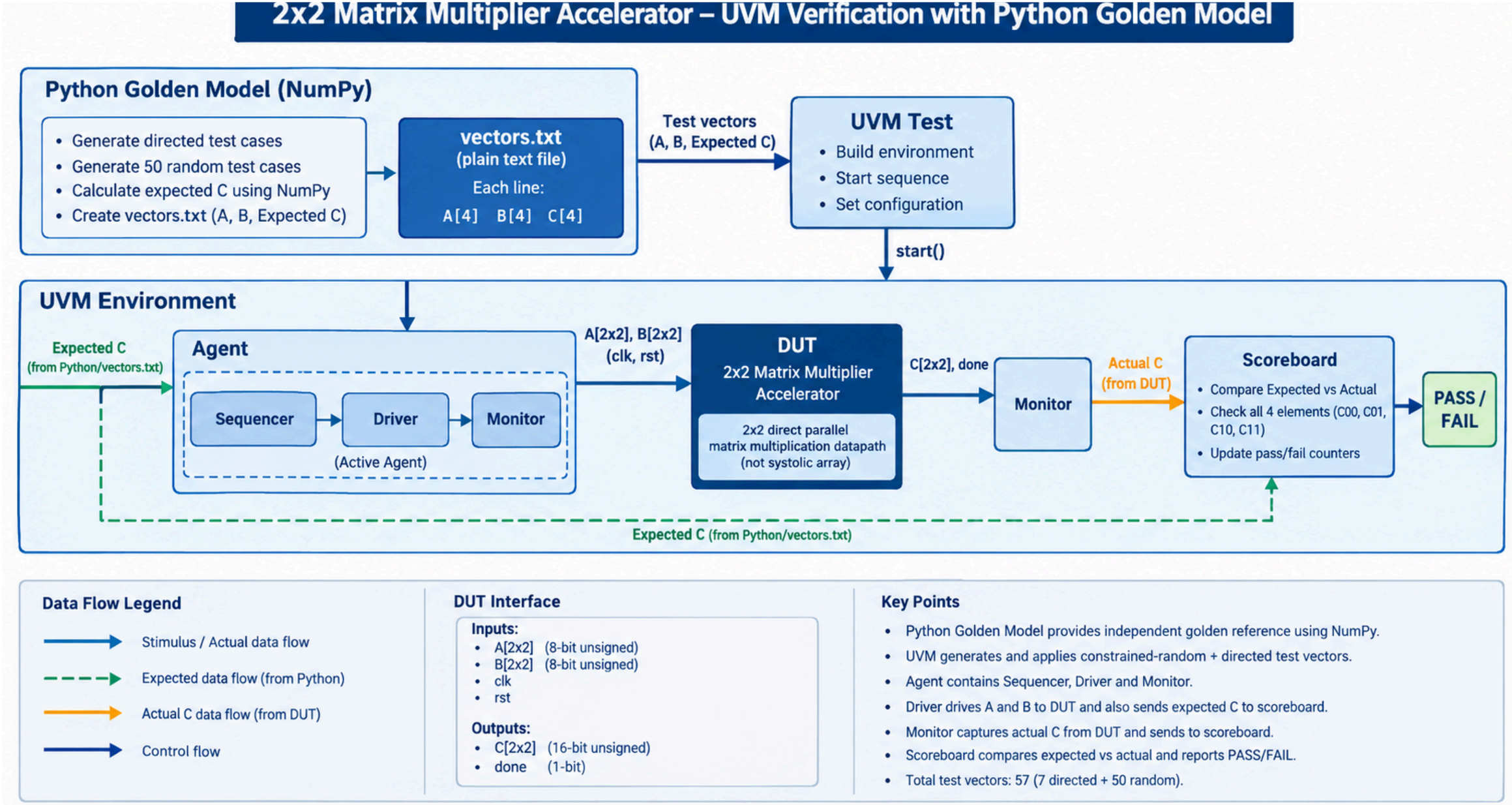
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Introduction

- Matrix multiplication is a key computation in AI and deep learning, but repeated operations can create a performance bottleneck in general-purpose processors.
- Our project implements a dedicated matrix multiplication accelerator using a systolic-array architecture, enabling parallel computation through multiple Processing Elements (PEs).
- The project has progressed from RTL design and basic simulation in Review 1 to systematic hardware verification using industry-standard UVM methodology.
- A Python/NumPy golden model generates test vectors and provides the expected matrix multiplication results as an independent reference.
- A SystemVerilog UVM testbench drives the test cases to the DUT, monitors the hardware outputs, and compares them with the golden results to ensure correctness across directed and randomized test cases.

Python – UVM verification environment



Project Progress till date

S.No.	Work Done
(1)	Development of the Python golden reference model using NumPy
(2)	Design of the complete UVM testbench architecture — transaction, sequence, driver, monitor, scoreboard, agent, environment, and test classes
(3)	Integration of the Python golden model with the UVM scoreboard; debugging and achieving a fully passing regression on Cadence NCSim

```
UVM_INFO /home/install/INCISIVE152/tools/methodology/UVM/CDNS-1.1d/sv/src/base/uvm_objection.svh(1268) @ 1186: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
UVM_INFO testbench.sv(212) @ 1186: uvm_test_top.env.sb [SB] TOTAL RESULT: 57 PASS, 0 FAIL

--- UVM Report catcher Summary ---


Number of demoted UVM_FATAL reports  :    0
Number of demoted UVM_ERROR reports  :    0
Number of demoted UVM_WARNING reports:    0
Number of caught UVM_FATAL reports   :    0
Number of caught UVM_ERROR reports   :    0
Number of caught UVM_WARNING reports :    0

--- UVM Report Summary ---

** Report counts by severity
UVM_INFO :   60
UVM_WARNING :    0
UVM_ERROR :    0
UVM_FATAL :    0
** Report counts by id
[RNTST]     1
[SB]    58
[TEST_DONE]     1
Simulation complete via $finish(1) at time 1186 NS + 44
```

Methodology overview with time line

- **Phase 1 - RTL Design (Weeks 1-4) DONE**

MAC unit -> 2x2 systolic array -> clocking, reset, done signal -> 8 directed tests passing

- **Phase 2 - NCSim Simulation (Weeks 5-6)DONE**

Compile in NCSim -> waveform verification in SimVision -> schematic verification (8 multipliers + 4 adders)

- **Phase 3 - Python Golden Model + UVM Testbench (Weeks 7-9) DONE**

NumPy golden model (57 vectors) -> UVM transaction/sequence/driver/monitor/scoreboard -> integrated and debugged -> 57 PASS, 0 FAIL on NCSim

- **Phase 4 - Coverage + SVA Assertions (Week 10) UPCOMING**

Define covergroups for edge cases -> embed SystemVerilog assertions in RTL -> achieve >90% functional coverage

- **Phase 5 - FPGA Synthesis and Deployment (Week 11) UPCOMING**

Synthesize in Xilinx Vivado -> collect LUT, DSP48, FF, Fmax -> program FPGA -> verify with ILA / UART

- **Phase 6 - AI Inference Demo + Final Report (Week 12) UPCOMING**

Demonstrate Output = Weight x Input on FPGA -> final report, GitHub repo, presentation

References

- ChipVerify (UVM Verification) - https://chipverify.com/tutorials/uvm
- Tutorial on NCSim – https://community.cadence.com/cadence_technology_forums/f/functional verification/35689/tutorial-on-ncsim